Lab 5
ITAI1371-Team BetaThetas
Reflective Journal: Data Preparation
Members: Brandy Gonzales, Derrick K. Gibbs-McGlaston, Victor Djuma, Pablo Lara, and Marlon Roberts

Brandy Gonzales

Before this lab, I didn't fully understand why Lab 4 mattered. In Lab 4 we explored the Titanic data and found the problems: Age had 177 missing values, Cabin was mostly empty, Fare had a few huge outliers over $500, and Sex and Embarked were words instead of numbers. Lab 5 helped me see what Lab 4 meant. Lab 4 was about finding the mess, and Lab 5 was about figuring out how to fix messy and missing data so a model can actually use it. Now I see these as two steps in the same workflow instead of two separate assignments. My biggest breakthrough was understanding mean versus median. What clicked for me was seeing the mean of five passengers' ages. Only one of them was high, but it pulled the mean higher than all the other numbers, which would give false data if I used it to fill in missing ages. The median uses the actual middle number in the list, so one outlier can't drag it around. That's why we filled the missing ages with the median. One-hot encoding also made more sense once I saw it in action. Embarked became Embarked_Q and Embarked_S, with 1s and 0s instead of letters. If we had just numbered the ports 1, 2, 3, the model would think one port was "bigger" than another, which isn't true. What confused me was when to use scaling and when not to. At first I thought scaling would help a decision tree, because I was thinking about how outliers skew data. After working through it, I realized it was the opposite. A decision tree only asks yes or no questions, like "Is Fare over 50?" Scaling shrinks the numbers but keeps them in the same order, so every passenger still lands on the same side of each split. That changed how I think about data prep: not every step is needed for every model, and I have to think about how the model makes its decisions. A question I still have is which models do need scaling, and how I'll know when to use it on my own projects.

Derrick K. Gibbs-McGlaston

My main contribution to the Module 05 Data Preparation lab was completing and reviewing the data preprocessing steps in the notebook.

I worked through the required tasks and checked the results after each step. This included handling the missing values in the Age column, converting the categorical Sex and Embarked information using one-hot encoding, and standardizing the Age and Fare features.

My focus was making sure each preparation step worked correctly before moving forward. I reviewed how the dataset changed after each operation instead of only checking whether the code ran without an error.

This contribution helped make sure the data preparation portion of the notebook was completed correctly and that the resulting data was in a better form for the next stages of the machine learning workflow.

Victor Djuma

For Lab 05, my contribution focused on completing the Data Preparation notebook in Google Colab and making sure each required task was completed correctly. I began by running the setup section, which loaded the Titanic dataset and displayed the missing values. This helped me identify that the dataset contained missing values, especially in the Age column, which was one of the main problems addressed in the lab. The notebook explained that raw data often requires preprocessing before it can be used by a machine learning model.

For the first task, I worked on handling missing values in the Age column. I calculated the median age and used it to replace the missing Age values. After completing the imputation, I verified the result by checking the number of missing values remaining in the Age column, which became zero. This task helped me understand how imputation can preserve useful data instead of simply deleting incomplete rows.

For the second task, I completed the One-Hot Encoding section. I used pd.get_dummies() to convert the categorical Sex and Embarked columns into numerical columns. I also used drop_first=True as directed in the notebook. After running the code, I checked the new DataFrame and confirmed that the original categorical columns had been replaced with encoded columns such as Sex_male, Embarked_Q, and Embarked_S.

For the third task, I completed the feature scaling section. I used StandardScaler from scikit-learn to scale the Age and Fare columns. After running the code, I reviewed the transformed values and confirmed that Age and Fare were standardized. This helped me understand why numerical features on different scales may need to be transformed before being used by some machine learning algorithms.

I also completed the three Knowledge Check questions at the end of the notebook. I explained why the median is useful for imputation, how One-Hot Encoding converts categorical information into numerical form, and why Decision Tree models generally do not require feature scaling.

Finally, I reviewed the notebook to make sure the required code cells executed successfully, saved the completed Jupyter notebook, and created a PDF version of the lab. My contribution to Lab 05 included completing the required coding tasks, checking the outputs, answering the Knowledge Check questions, and preparing my completed work for submission.

Pablo Lara

Going into this lab, I was intimidated. I know working with data can be a huge task, and I wasn't sure where to start. The Titanic dataset had 891 passengers, but the first output showed 177 missing ages, 687 missing cabin values, and 2 missing embarkation ports. Seeing that made it clear why data preparation matters: a model can't learn from blanks or from text. Once I started working through each problem step by step, I began to gain confidence.

The first real challenge was figuring out how to turn "male" and "female" into numbers. Models only understand math, so text has to be converted. I thought about Boolean variables and how they split into exactly two directions, True or False. I took a risk and wondered if the same idea would work for male and female, and it paid off. With one-hot encoding and drop_first=True, the Sex column became a single column, Sex_male, where 1 means male and 0 means female. A separate female column isn't needed, because if a passenger isn't male, the data already tells you they're female. When I looked at the output, pandas actually displayed the column as True/False, which confirmed my thinking. The same idea applied to the Embarked column, where giving the ports numbers like 0, 1, and 2 would have created an order that doesn't exist.

The most memorable moment was a cell that ran when it shouldn't have. Almost every code cell in the notebook gave me a syntax error, because the file had lost its line breaks and every statement was squashed onto one line. At first I thought it was an indentation problem. When the scaling cell ran without any error, I was surprised, and it made me feel something was wrong. When I reviewed it, I realized the code after the first # was being treated as a comment, so the only thing that actually ran was the import line. Nothing had been scaled. I corrected it, finished the task, and felt more confident in my work afterward. The lesson I'm taking away is that "no error" does not mean "it worked." From now on I will check the output to make sure the code did what I expected.

What surprised me most overall was how difficult it can be to come up with clever ways to transform data so it fits what a model needs. Before this lab, I thought cleaning data was mostly deleting bad rows. Instead, every column needed its own decision: filling missing ages with the median, encoding categories as 0/1 columns, and scaling Age and Fare so that Fare's larger numbers wouldn't seem more important.

This lab also connected to what I already knew. My Python class helped me understand the code, and seeing pandas and NumPy again after Lab 2 reminded me how important they are in machine learning. They aren't one-time tools; they show up at every step of the workflow.

AI disclosure: I used Claude to explain concepts and to help me understand why the python code kept failing.

Marlon Roberts

For this lab I contributed to doing the lab, learning about data preparation and how its a very important aspect of cleaning a dataset. In this lab we loading the raw dataset into the environment and analyzing its dimensions to understand the scope of the data. I resolved structural issues, handling missing entries, and removing noise to prevent skewed models. Preparing non-numeric and continuous variables so mathematical algorithms can interpret them. After that we split the data up to train it.